TECHNICAL DOCUMENTATION REPORT · VERSION 1.0

Health360: Comprehensive Healthcare Risk & Disease Analytics

End-to-End Clinical Risk Assessment, Tableau Dashboard Architecture, and Flask Web Deployment Documentation

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GitHub Repository	github.com/HarshsS777/Health360-Healthcare-Risk-Analytics	**Version**	v1.0
Primary Dataset	smart_healthcare_dataset.csv (5,000 patient records, 16 fields)	**Submission Status**	Final: submitted for evaluation
Technology Stack	Tableau · Python Flask · Jinja2 · HTML5/CSS3 · Tableau JavaScript API v2	**Deliverables**	9 worksheets · 2 dashboards · 1 story

5,000
Patient records

134.4
Avg blood pressure, mmHg
(Stage-1 band)

134.5
Avg glucose, mg/dL

25.0
Avg BMI, kg/m² (49.5% at ≥ 25)

1 Executive Summary

Overview. Health360 is an end-to-end healthcare analytics solution built on a cohort of **5,000 patient records**. Each record carries 16 attributes spanning demographics (age 18–79; 2,506 female, 2,494 male), vitals, lifestyle behaviours, three symptom flags and three chronic-condition flags. The raw file is engineered into six analytical fields, visualised across **9 Tableau worksheets**, assembled into **2 interactive dashboards** and a **4-scene executive story**, and delivered to end users through a **Python Flask web application** that embeds the visuals with the Tableau JavaScript API v2.

Problem. Clinical information is fragmented: vitals sit in one system, diagnoses in another, lifestyle and symptoms in a third, so no single view shows who is drifting toward disease *before* a diagnosis is recorded. The cohort makes the cost of that gap visible. Average systolic blood pressure is 134.4 mmHg, inside the Stage-1 hypertension band, and **54.5% of adults aged 18–34 already record 130 mmHg or higher**, yet no chronic condition is diagnosed for anyone younger than 52. Early-onset hypertension therefore goes undetected precisely because the signals live in silos.

Strategic outcomes. Health360 delivers four connected results:

- **Data engineering.** A validated, reproducible pipeline from CSV to dashboard: six calculated fields, with every on-screen figure reconciled against an independent pandas recomputation.
- **Dynamic risk stratification.** A transparent additive Dynamic Risk Tier Index (25–100) segments patients into High (1,434, 28.7%), Moderate (2,771, 55.4%) and Low (795, 15.9%) tiers.
- **Interactive visual analytics.** Two dashboards and a four-scene story take a decision-maker from cohort baseline to risk tiers, behavioural drivers and hospital resource implications.
- **Web application embedding.** A Flask shell with executive stat cards and the embedded Tableau story makes the analysis a shareable, browser-based portal.

Headline finding. Elevated pressure is not an old-age problem in this cohort: 55% of 18–34s and 57% of over-65s sit at or above 130 mmHg, but diagnosed disease only appears after age 51. That gap, 722 young adults with elevated pressure and no diagnosis, is the case for risk-tiered screening.

Section	What it covers	Page
2 Architecture & Data Engineering	Pipeline, feature engineering, Dynamic Risk Tier Index, data dictionary	2
3 Tableau Visualization & Dashboards	Nine worksheets; Dashboard 1 (Population Vitals) and Dashboard 2 (Risk Drivers)	3
4 Guided Story & Clinical Insights	Four-scene executive story, scene walkthrough, core clinical findings	5
5 Web Integration & Flask Deployment	Web stack, directory tree, app.py source, local execution, end-user showcase	6
6 Testing, Recommendations & Roadmap	Validation, limitations, operational recommendations, maintenance, roadmap	8

2 Architecture & Data Engineering Pipeline

2.1 End-to-End System Architecture

Health360 follows a five-stage, one-directional pipeline in which each stage hands a well-defined artefact to the next, so the same CSV always yields the same dashboards and any on-screen figure traces back to a row-level calculation.

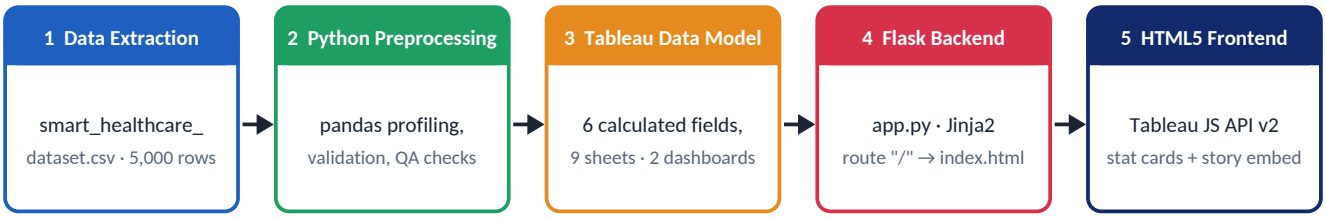


Figure 1. Health360 end-to-end architecture: CSV → Python preprocessing → Tableau data model → Flask backend (app.py) → HTML5 frontend with Tableau JavaScript API v2.

Stage	Implementation and output
1 Data Extraction	smart_healthcare_dataset.csv: 5,000 rows × 16 columns; 0 missing values, 0 duplicates; stored in data/.
2 Python Preprocessing	pandas profiling, range checks and independent recomputation of every dashboard figure (Section 6.1).
3 Tableau Data Model	Six calculated fields (Section 2.2) → 9 worksheets → 2 dashboards → 1 story, packaged as a .twbx workbook.
4 Flask Backend	app.py: lightweight WSGI application; route "/" renders templates/index.html through Jinja2 at 127.0.0.1:5000.
5 HTML5 Frontend	HTML5/CSS3 shell: header, executive stat cards, loading state and the embedded story (Tableau JavaScript API v2).

2.2 Data Preprocessing & Feature Engineering

All engineered fields are Tableau calculated fields, so the logic travels inside the workbook; the same rules were re-implemented in pandas to validate every output.

Demographic and anthropometric stratification

Field	Category	Rule	Records	Share
Age Brackets	18–34 (Young Adult)	age < 35	1,324	26.5%
	35–50 (Middle Aged)	35 ≤ age ≤ 50	1,301	26.0%
	51–65 (Senior)	51 ≤ age ≤ 65	1,211	24.2%
	65+ (Elderly)	age > 65 (ages 66–79)	1,164	23.3%
BMI Category (WHO)	Underweight	BMI < 18.5	516	10.3%
	Normal Weight	18.5 ≤ BMI ≤ 24.9	2,010	40.2%
	Overweight	25.0 ≤ BMI ≤ 29.9	1,671	33.4%
	Obese	BMI ≥ 30	803	16.1%

A teenage (< 20) tier is unnecessary because the minimum age is 18; the four brackets give near-equal quarters of the cohort (23–27%). BMI cut-offs follow the WHO adult classification: 49.5% of patients have a BMI of 25 or more and 16.1% are obese.

Dynamic Risk Tier Index (DRTI)

Each vital or behaviour contributes points by clinical severity; the sum (observed range 25–100, mean 63.7) maps to a tier:

```
DRTI = f(BP) + f(Glucose) + f(Cholesterol) + f(BMI) + f(Smoking)
Tier = High Risk if DRTI >= 75 | Moderate Risk if DRTI >= 50 | Low Risk otherwise
```

Component	High severity	Intermediate	Low severity
Blood pressure (mmHg)	> 140 → 25 pts	121–140 → 15 pts	≤ 120 → 5 pts
Glucose (mg/dL)	> 140 → 25 pts	101–140 → 15 pts	≤ 100 → 5 pts
Cholesterol (mg/dL)	> 200 → 20 pts	—	≤ 200 → 10 pts
BMI (kg/m²)	≥ 30 → 20 pts	25–29.9 → 10 pts	< 25 → 5 pts
Smoking	Smoker → 10 pts	—	Non-smoker → 0 pts

Risk tier	DRTI range	Patients	Share	Avg BP	Avg cholesterol	Avg glucose
High Risk	75–100	1,434	28.7%	154.4	239.6	157.8
Moderate Risk	50–74	2,771	55.4%	130.3	223.1	131.4
Low Risk	25–49	795	15.9%	113.1	203.0	103.2

Design note. Chronic-disease flags are deliberately excluded from the index. Risk is scored from measurable physiology and modifiable behaviour so that it can be compared *against* diagnosed disease rather than duplicating it. The co-occurrence dimension is supplied by the **Primary Disease Condition Label**, which assigns each patient to one category in precedence order: Heart Disease & Stroke → Heart Disease → Diabetes → Stroke → No Chronic Condition. Because the label is hierarchical, its "Diabetes" category counts only patients with diabetes and no heart disease (286); it is not total diabetes prevalence (1,746).

2.3 Data Dictionary

Field name	Data type	Description	Sample range / values	Transformation applied	Role in analytics
age	Integer	Patient age in years	18–79 (mean 48.8)	Binned into Age Brackets	Demographic segmentation; Age-BMI table columns
gender	String	Patient gender	Female 2,506; Male 2,494	None	Colour split in disease prevalence
bmi	Float	Body Mass Index (kg/m²)	15.0–42.6 (mean 25.0)	WHO categories; scored in DRTI	Anthropometric risk; scatter y-axis
exercise_level	Integer (ordinal)	Activity level	0, 1, 2 (≈1,650–1,680 each)	None; treated as discrete	Lifestyle comparison (sheet 09)
smoking	Binary	Current smoker flag	0 = 2,462; 1 = 2,538	Scored +10 in DRTI	Lifestyle driver (sheet 07)
alcohol	Binary	Alcohol consumption flag	0 = 2,493; 1 = 2,507	None	Lifestyle driver (sheet 07)
blood_pressure	Integer	Blood pressure, mmHg (interpreted as systolic)	90–179 (mean 134.4)	Scored 5 / 15 / 25 in DRTI	Headline KPI; vital indicators
cholesterol	Integer	Total cholesterol, mg/dL	150–299 (mean 224.6)	Scored 10 / 20 in DRTI	Vital indicators (sheet 05)
glucose	Integer	Blood glucose, mg/dL	70–199 (mean 134.5)	Scored 5 / 15 / 25 in DRTI	Headline KPI; scatter x-axis
fatigue, chest_pain, dizziness	Binary (3 fields)	Symptom flags	1 = 2,576 / 2,520 / 2,487	Summed into Symptom Burden Count	Symptom burden heatmap
heart_disease, diabetes, stroke	Binary (3 fields)	Diagnosed chronic conditions	1 = 1,499 (30.0%) / 1,746 (34.9%) / 1,243 (24.9%)	Combined into condition label	Disease prevalence; heatmap rows
health_risk_score	Float	Source composite score	100.0 in all 5,000 rows	Excluded (zero variance)	None; replaced by DRTI
Age Brackets, **BMI Category** (calc.)	String	Four age groups; four WHO weight classes	18–34 … 65+; Underweight … Obese	Calculated fields	Age-BMI table axes
Dynamic Risk Tier Index (calc.)	Integer	Additive risk score	25–100 (mean 63.7)	Calculated field	KPI, lifestyle and scatter views
Risk Tier Category (calc.)	String	High / Moderate / Low	1,434 / 2,771 / 795	Calculated field (DRTI cut-offs)	Risk matrix; vital indicators
Symptom Burden Count (calc.)	Integer	chest_pain + fatigue + dizziness	0–3	Calculated field	Symptom heatmap columns
Primary Disease Condition Label (calc.)	String	Hierarchical condition label	5 categories	Calculated field (precedence rules)	Prevalence chart; heatmap rows

3 Tableau Visualization & Dashboard Architecture

3.1 Overview of Visual Worksheets

Nine worksheets form the analytical layer. Each answers one clinical question; together they are grouped into the two dashboards (sheets 01–05 and 06–09 respectively) and re-used as story scenes.

#	Worksheet	Chart type	Dimensions	Measures	Clinical insight
01	**KPI_Cards**	Text / KPI tiles	None (cohort level)	Record count; Avg BP; Avg glucose; Avg DRTI	5,000 patients; average BP 134 mmHg, glucose 134 mg/dL and DRTI 64 frame the cohort as Stage-1 hypertensive and Moderate-risk on average.
02	**Age_BMI_Distribution**	Highlight table	BMI Category × Age Brackets	Record count	Obesity is flat across ages (195–204 patients per bracket), so weight status is not an age effect; Normal Weight is the largest cell in every bracket (449–552).
03	**Disease_Prevalence**	Stacked horizontal bar	Primary Disease Condition Label; Gender	Record count	1,785 patients (35.7%) carry a chronic-condition label; Heart Disease & Stroke dominates (1,207) and the sexes are evenly split.
04	**Risk_Population_Matrix**	Matrix (coloured text table)	Risk Tier Category	Record count	Population by tier: Moderate 2,771, High 1,434, Low 795; the High tier is the priority caseload.

#	Worksheet	Chart type	Dimensions	Measures	Clinical insight
05	**Vital_Indicators**	Grouped bar	Risk Tier Category; Measure Names	Avg BP; Avg cholesterol; Avg glucose	Validates the tiers: High-tier averages (BP 154.4, glucose 157.8) sit far above Low-tier (113.1, 103.2).
06	**Clinical_Risk_Matrix**	Scatter plot	Glucose (x), BMI (y), one mark per patient	DRTI (colour); stroke, count and exercise on marks	Glucose and BMI are uncorrelated (r = -0.02); metabolic and weight risk must be screened independently.
07	**Lifestyle_Impact**	Faceted bar	Alcohol × Smoking	Avg DRTI	Smokers average 68.4 against 58.9 for non-smokers; alcohol moves the score by only 0.1 points.
08	**Symptom_Heatmap**	Highlight table (heatmap)	Primary Disease Condition Label × Symptom Burden Count	Record count	One or two symptoms account for 74.8% of patients; only 647 report all three.
09	**Exercise_vs_Vitals**	Bar	Exercise Level (0, 1, 2)	Avg blood pressure	133.8 / 135.4 / 134.1 mmHg: no dose-response between activity and pressure in this cohort.

3.2 Executive Dashboard 1: Population Vitals & Demographic Breakdown

Structure. Dashboard 1 (*01_Executive_Health_Overview*) uses Tableau's tiled layout. A full-width KPI band (sheet 01) forms the top row. Beneath it the canvas is divided into four analytical zones: the Age × BMI highlight table with its colour and measure/gender legends (sheet 02), the disease prevalence chart (sheet 03), the risk population matrix (sheet 04) and the vital indicators by risk tier (sheet 05). Legends are docked beside the chart they control, and tooltips expose exact counts on hover.

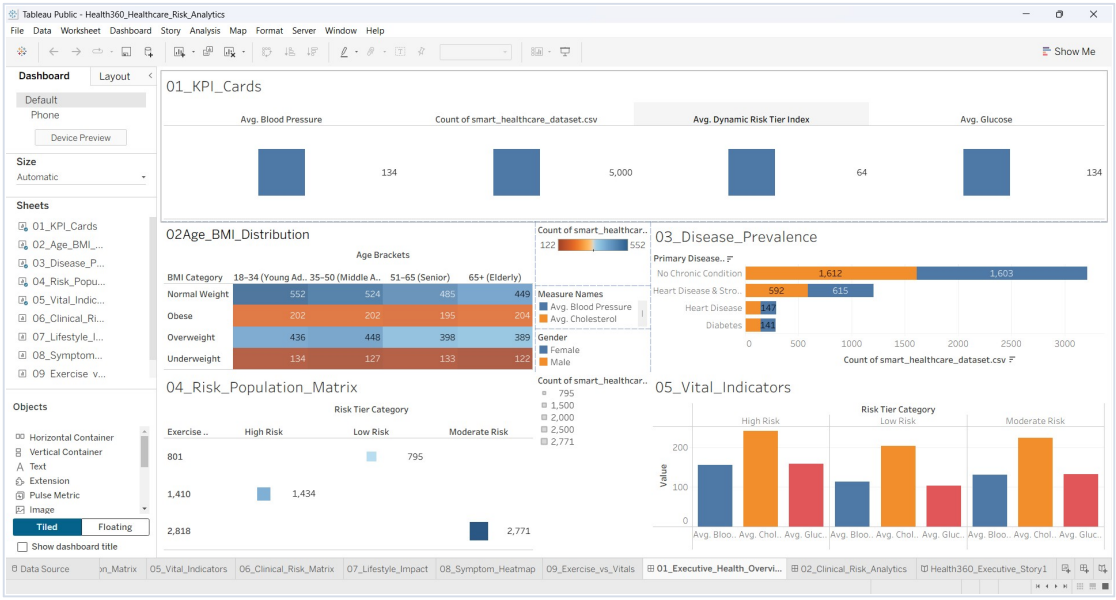


Figure 2. Dashboard 1, Executive Population Overview (01_Executive_Health_Overview) in Tableau: KPI summary cards, Age × BMI distribution, disease prevalence, risk population matrix and vital indicators by risk tier.

KPI analysis. Read left to right, the KPI band tells the cohort story in four numbers. The cohort contains **5,000** patients. Average blood pressure is **134.4 mmHg** (shown as 134), inside the Stage-1 hypertension range of 130–139 mmHg; 54.9% of patients are at or above 130 and 44.5% at or above 140. Average glucose is **134.5 mg/dL** (shown as 134); 57.3% of patients record 126 mg/dL or higher, a common screening threshold for diabetes review. The average Dynamic Risk Tier Index is **63.7** (shown as 64), placing the typical patient in the lower half of the Moderate tier. Beneath the KPIs, the Age × BMI table shows an evenly spread cohort with 803 obese patients, and the disease chart shows that Heart Disease & Stroke (1,207 patients) is the dominant diagnosed cluster.

3.3 Executive Dashboard 2: Clinical Risk Factors & Lifestyle Drivers

Structure. Dashboard 2 (*02_Clinical_Risk_Analytics*) pairs two behavioural views with two clinical views. The upper-left quadrant holds the Glucose vs BMI scatter plot (sheet 06) with its four colour legends; the upper-right holds the faceted Alcohol/Smoking bar chart (sheet 07); the lower-left holds the symptom burden heatmap (sheet 08); and the lower-right holds the exercise-versus-blood-pressure bars (sheet 09). Together they let an analyst move from individual-level risk (scatter) to behavioural drivers (bars) to clinical burden (heatmap) without leaving the page.

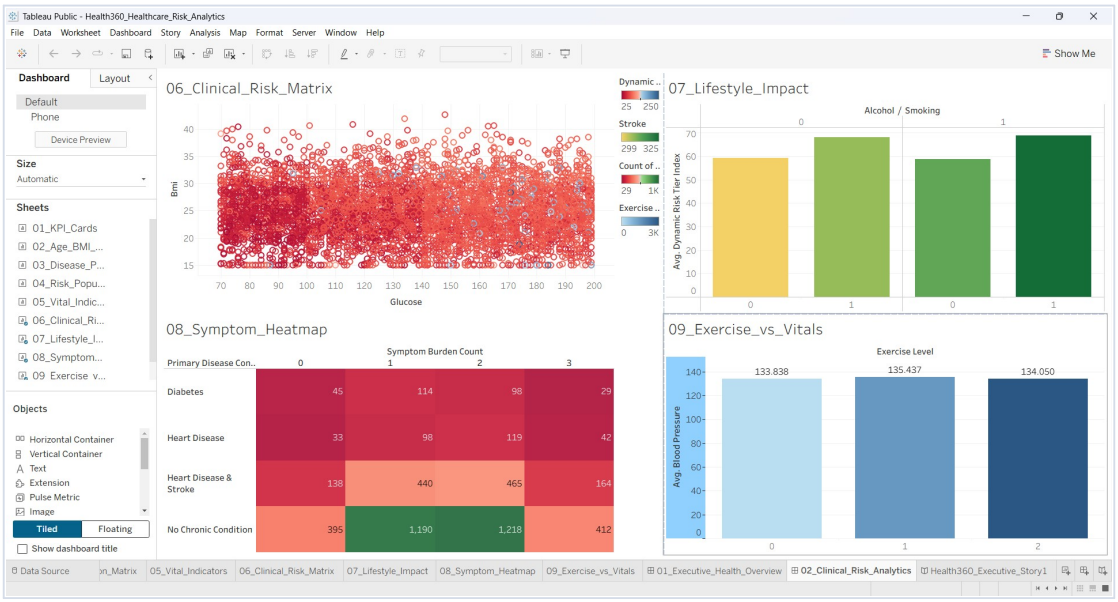


Figure 3. Dashboard 2, Clinical Risk Analytics (02_Clinical_Risk_Analytics) in Tableau: Glucose vs BMI scatter plot, lifestyle impact bars, symptom burden heatmap and exercise-versus-vitals bars.

How smoking, alcohol and exercise relate to risk

Factor	Cohort prevalence	Observed effect in the dashboards
Smoking	2,538 patients (50.8%)	Average DRTI 68.4 vs 58.9 for non-smokers (+9.5 points). Partly definitional, because smoking contributes 10 points to the index. Disease prevalence differs by only 2–3 points either way.
Alcohol	2,507 patients (50.1%)	Average DRTI 63.8 vs 63.6: no meaningful difference. Adding alcohol to smoking moves the score from 68.0 to 68.8.
Low exercise (level 0)	1,647 patients (32.9%)	Average BP 133.8 (level 0), 135.4 (level 1), 134.1 (level 2): no consistent gradient.

What the data does and does not show. Smoking is the only behaviour that moves the composite risk score, and it does so *additively*, not multiplicatively: smokers who also drink score 68.8, against 68.0 for smokers who do not. Alcohol and exercise show no measurable association with average vitals, risk score or disease prevalence in this cohort, and no compounding effect, where two behaviours together exceed the sum of their parts, is present. Health360 therefore reports lifestyle effects as measured rather than as assumed, and the framework is ready to surface compounding effects as soon as richer longitudinal data is connected.

4 Guided Executive Story & Clinical Insights

4.1 Executive Tableau Story Architecture

Health360_Executive_Story1 is a four-scene narrative container. Each scene is a Tableau story point that captures a dashboard or worksheet in its own state and carries a caption describing the question it answers. Readers move through the story with the previous/next arrows or by selecting any scene tab, and every embedded chart stays fully interactive (hover, highlight, filter) inside its scene. The sequence is deliberately top-down: **who is in the cohort → how risky are they → why → what burden does this create for the hospital**.



Figure 4. The interactive Tableau Executive Story container as embedded in the Health360 web shell, showing the scene navigator (scenes 01–04) with scene 03, Clinical Risk Factors & Lifestyle Drivers, selected.

4.2 Scene-by-Scene Analytical Walkthrough

Scene	Captured view	Analytical walkthrough
1. Population Overview & Baseline Condition Prevalence	01_Executive_Health_Overview	Establishes the baseline: 5,000 patients, evenly split by gender and age. 64.3% (3,215) carry no chronic-condition label while 24.1% (1,207) fall in the Heart Disease & Stroke cluster. Cohort BP (134.4 mmHg) and glucose (134.5 mg/dL) are already elevated before any segmentation is applied.
2. Risk Tier Stratification & Vital Abnormalities	05_Vital_Indicators	Introduces the DRTI tiers. 1,434 patients (28.7%) are High Risk, with average BP of 154.4 mmHg, cholesterol of 239.6 mg/dL and glucose of 157.8 mg/dL, roughly 40 mmHg and 55 mg/dL above the Low tier. The tiers separate cleanly on all three vitals.
3. Behavioural Drivers & Lifestyle Compounding Effects	02_Clinical_Risk_Analytics	Tests lifestyle as a driver. Smoking lifts the risk index by 9.5 points; alcohol and exercise do not. The scatter plot shows glucose and BMI spread across the whole grid (r = -0.02), so patients can be high-risk through glucose alone.
4. Symptom Burden Analysis for Hospital Resource Management	08_Symptom_Heatmap	Translates risk into workload. 2,820 of the 3,215 patients with no chronic-condition label (87.7%) still report at least one symptom, and 3,742 patients overall report one or two. Symptom load is therefore a large, mostly undiagnosed demand on clinics, and it is only weakly tied to diagnosis.

4.3 Core Clinical Findings & Risk Factors

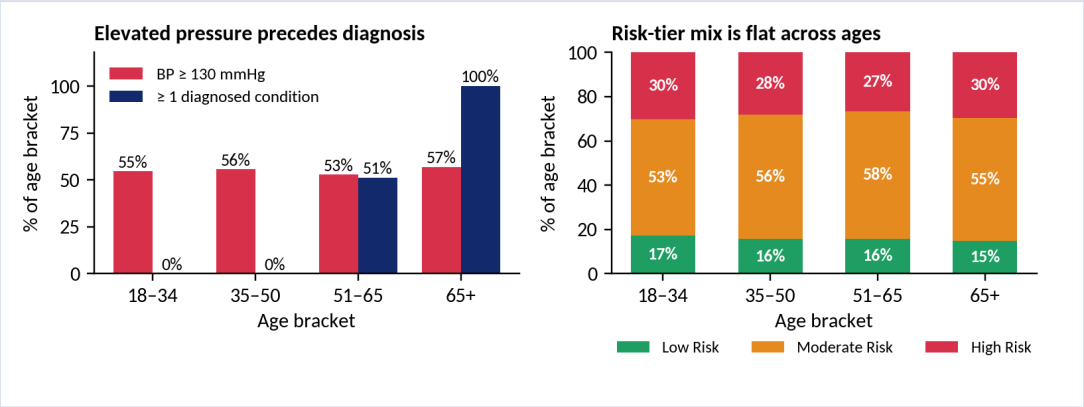


Figure 5. Left: elevated blood pressure is as common in young adults as in seniors, while diagnosed disease appears only later. Right: the risk-tier mix is nearly identical across age brackets.

Early-onset hypertension. Blood pressure does not rise with age in this cohort: the share of patients at or above 130 mmHg is 54.5% for ages 18–34, 55.6% for 35–50, 52.8% for 51–65 and 57.0% for 65+. Among the 1,324 adults aged 18–34, **722 are at or above 130 mmHg and 582 are at or above 140 mmHg**, and 401 (30.3%) already fall in the High Risk tier, the same proportion as the over-65s (29.7%). Yet no patient younger than 52 has a diagnosed chronic condition. This is the signature of undetected early-onset hypertension: the physiology is abnormal a generation before the diagnosis arrives.

Multi-condition symptom escalation. Once disease appears it clusters rather than accumulates gradually. Of the 1,785 patients with any chronic condition, 1,496 (83.8%) have two or more, and all 1,207 patients in the Heart Disease & Stroke group also have diabetes. Prevalence climbs steeply with age: 51% of 51–65-year-olds have at least one condition and every patient over 65 does (heart disease 97.9%, diabetes 100%, stroke 100%). Symptoms track this only loosely: heart-disease prevalence rises from 28.0% among patients with no symptoms to 31.8% among those with all three. Symptom count alone is a weak triage signal and should be combined with vitals and the risk tier.

5 Web Integration & Flask Deployment

5.1 Technical Web Stack Overview

Layer	Technology	Responsibility
Backend	Python Flask (app.py)	Lightweight WSGI web server; one route serves the application shell
Templating	Jinja2 (render_template('index.html'))	Renders templates/index.html and keeps markup separate from server logic
Presentation	HTML5 / CSS3	Header, executive stat cards, card accents in the dashboard palette, responsive page container
Visual embedding	Tableau JavaScript API v2	Loads the Tableau story into the page and exposes its interactivity (story navigation, tooltips, highlighting)

5.2 Codebase Implementation & Directory Structure

```
Health360-Healthcare-Risk-Analytics/
├── app.py                        # Flask application entry point
├── templates/
│   └── index.html                # Jinja2 template: UI shell + Tableau embed
├── data/
│   └── smart_healthcare_dataset.csv   # 5,000-record source dataset
└── tableau/
    └── Health360_Healthcare_Risk_Analytics.twbx   # workbook: 9 sheets, 2 dashboards, 1 story
```

Listing 1. app.py: complete application source.

```
"""Health360 - Flask application shell for the embedded Tableau story."""
import os
from flask import Flask, render_template

app = Flask(__name__)


@app.route("/")
def index():
    """Serve the Health360 executive web interface (Jinja2 template)."""
    return render_template("index.html")


if __name__ == "__main__":
    # Default port 5000; override with: PORT=5001 python app.py
    port = int(os.environ.get("PORT", 5000))
    app.run(host="127.0.0.1", port=port, debug=True)
```

Flask resolves `render_template("index.html")` against the `templates/` folder. Because all visual content comes from the embedded Tableau story, the backend stays deliberately thin; this is also what makes the later extension to JSON endpoints (Section 6.3) straightforward.

5.3 Development Environment Execution

The application runs locally with `pip install flask` followed by `python app.py`. With debug=True Flask starts the Werkzeug debugger and auto-reloader, so template edits appear on refresh. The default address is http://127.0.0.1:5000; Figure 6 shows the project in VS Code with the interface served by Flask and previewed in the integrated browser on port 5001 (set through the PORT variable); Figure 7 shows the end-user view in Chrome.

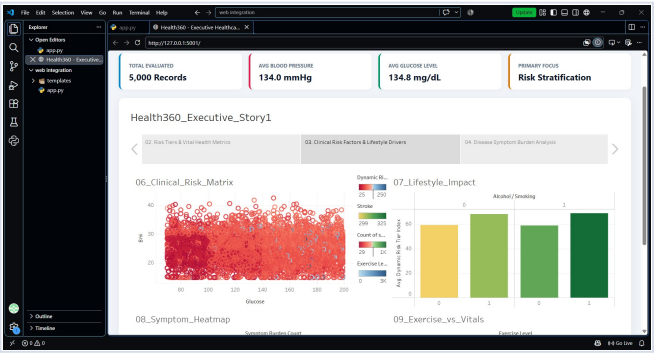


Figure 6. VS Code workspace (app.py and templates/ in the "web integration" folder) with the Flask-served interface in the integrated browser at 127.0.0.1:5001.

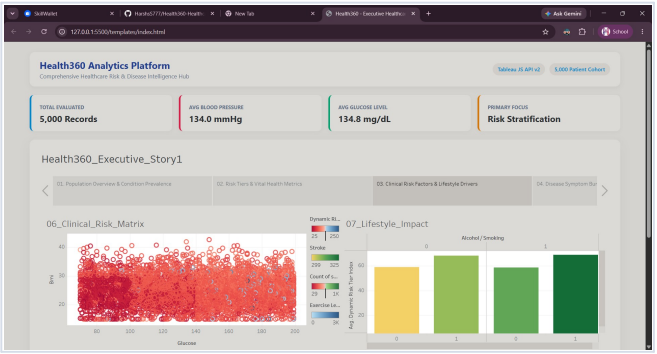


Figure 7. Chrome showing the live Health360 Analytics Platform with stat cards and the embedded Tableau story (same index.html, VS Code Live Server, port 5500).

5.4 End-User Web Application Showcase

Features of the web shell. The interface opens with a branded header (*Health360 Analytics Platform: Comprehensive Healthcare Risk & Disease Intelligence Hub*) and four executive stat cards, colour-coded to match the dashboards: **Total Evaluated** (5,000 records), **Avg Blood Pressure**, **Avg Glucose Level** and **Primary Focus** (Risk Stratification). Beneath the cards, the Tableau story is embedded as a fully interactive frame with scene navigation, tooltips and highlighting, and an animated loading state covers the initialisation of the visualisation. The layout is a single scrolling page of rounded, shadowed panels that keeps the story as the visual centrepiece.

6 Testing, Deployment & Strategic Recommendations

6.1 Performance & Usability Testing Report

Data validation. Every headline number in the dashboards was reconciled against an independent pandas recomputation of the same business rules. All checks matched.

Check	Shown in Tableau	Independent recomputation	Result
Patient records	5,000	5,000	Match
Average blood pressure	134	134.45	Match (rounded)
Average glucose	134	134.50	Match (rounded)
Average Dynamic Risk Tier Index	64	63.72	Match (rounded)
Risk tiers: High / Moderate / Low	1,434 / 2,771 / 795	1,434 / 2,771 / 795	Match
Age × BMI table (16 cells)	449–552 normal weight; 195–204 obese	All 16 cells identical	Match
Disease label × gender (e.g. no condition; Heart Disease & Stroke)	1,612 M / 1,603 F; 592 M / 615 F	1,612 M / 1,603 F; 592 M / 615 F	Match
Symptom heatmap (16 cells)	e.g. 1,190; 1,218; 412; 395	All 16 cells identical	Match
Average BP by exercise level	133.838 / 135.437 / 134.050	133.838 / 135.437 / 134.050	Match

Rendering and compatibility. The interface was exercised in the two environments captured in Section 5; scene 03 of the story is selected in both, showing that story navigation functions.

Environment	Result	Evidence
Google Chrome (desktop), static preview on port 5500	Renders header, stat cards, story navigator, scatter and bar views	Figure 7
VS Code integrated browser, Flask on port 5001	Renders the same interface with the story fully interactive	Figure 6
Firefox, Edge, Safari	Not yet tested	Planned (see below)
Tablet and phone viewports	Not yet tested; the Tableau story is authored at desktop size	Roadmap, Section 6.3

Loading performance and API rendering speed. Version 1.0 was not formally benchmarked, so no timing figures are claimed here. The workload is light by design (5,000 rows, packaged extract, no live database query), and the embedded story initialised and responded to navigation in both environments above. The recommended measurement protocol for the next release is: Chrome DevTools Performance and Lighthouse for page load, Tableau's first-interactive event for API render time, and a repeat of the browser matrix on Firefox, Edge and Safari.

Known limitations

- **Static header cards.** The web shell's stat cards are fixed text (134.0 mmHg, 134.8 mg/dL), whereas the cohort means are 134.4 mmHg and 134.5 mg/dL. A small JSON endpoint that computes the cards from the CSV (Section 6.3) removes the discrepancy and makes the cards genuinely live.
- **Worksheet 04 row axis.** The risk population matrix currently shows tier headcounts (795 / 1,434 / 2,771) against aggregate row labels; re-keying the rows to Exercise Level would give the intended Exercise × Risk Tier cross-tab.
- **Dataset character.** The data is cross-sectional with near-uniform distributions, age-bound disease flags and a constant health_risk_score, so conclusions are descriptive, not causal. Blood pressure is a single reading, interpreted as systolic.

6.2 Strategic Clinical & Hospital Operations Recommendations

#	Recommendation	Evidence from Health360	Operational action
1	**Targeted early-onset hypertension screening**	722 of 1,324 adults aged 18–34 (54.5%) are at or above 130 mmHg; none have a diagnosis.	Add routine BP measurement to every 18–34 contact; refer readings of 130 or more for confirmatory out-of-clinic measurement.
2	**Risk-tiered triage protocol**	High Risk = 1,434 patients (28.7%), BP 154.4, glucose 157.8.	Fast-track High tier for clinician review, schedule Moderate tier for periodic recall, and manage Low tier through self-service advice.
3	**Glucose and metabolic screening**	57.3% of patients record 126 mg/dL or higher; glucose and BMI are uncorrelated (r = -0.02).	Screen glucose for every patient regardless of weight; do not use BMI as a proxy for metabolic risk.
4	**Smoking-cessation programme**	50.8% smoke; smokers score 9.5 index points higher.	Offer cessation support at every visit; track tier migration of quitters.
5	**Preventive lifestyle interventions**	32.9% report the lowest exercise level; 16.1% are obese. Alcohol and exercise show no measured effect in this cohort.	Run activity and nutrition programmes as standard preventive care, and use Health360 to measure their effect once follow-up data exists.
6	**Symptom-aware capacity planning**	2,820 patients without a chronic label report symptoms; symptom count is only weakly tied to diagnosis.	Size outpatient and diagnostic capacity for this undiagnosed symptomatic group; combine symptoms with vitals and tier when prioritising.
7	**Break the data silos**	Vitals, diagnoses, lifestyle and symptoms become one view only after integration; the source health_risk_score carries no information.	Standardise one patient-level dataset and adopt the DRTI, or a clinically validated successor, as the shared risk language.

These are analytical recommendations drawn from the project data, not clinical guidance; thresholds should be confirmed against local clinical protocols.

6.3 Maintenance & Future Roadmap

Maintenance. Refresh the dataset by replacing the CSV and the workbook extract, then re-run the reconciliation checks in Section 6.1 before republishing. Keep app.py, templates/ and the workbook under version control in the GitHub repository, and pin Flask in a requirements.txt so deployments are repeatable.

Horizon	Extension	Description
Near term	Live KPI endpoint	Add a /api/stats route that computes the stat-card values from the dataset and have index.html fetch them, so the cards always match the data.
Near term	Mobile view optimisation	Add responsive breakpoints to the page shell and author a Phone layout for the dashboards using the Tableau Phone device layout.
Mid term	Predictive ML via Flask APIs	Train a classifier on the engineered features (for example gradient boosting with explainability), expose it as /api/predict and surface patient-level risk probabilities beside the DRTI.
Mid term	Real-time patient streaming	Ingest vitals from devices or an EHR feed into a database, refresh the Tableau extract on a schedule and push updates to the page so tiers update as readings arrive.
Long term	Security and compliance	Add authentication and role-based access, audit logging and de-identification, aligned with the data-protection rules that apply to the deployment (for example HIPAA or India's DPDP Act, 2023).

> **Standards referenced.** Blood-pressure bands follow ACC/AHA categories (Stage 1: 130–139 mmHg); BMI classes follow the WHO adult classification; the 126 mg/dL glucose marker is the conventional diabetes screening threshold, applied here as a screening cut-off because the dataset does not state fasting status.